

Anisha Mohandass

Data Scientist | NLP & Generative AI Engineer

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Location: Puducherry, India

Professional Summary

Data Scientist with **3.10 years** of experience in Natural Language Processing (NLP) and building production-grade language solutions. Skilled in Python, SQL, Machine Learning, and Generative AI concepts. Experienced in working with structured and unstructured data, building NLP pipelines, and optimizing model performance. Currently upskilling in Generative AI and Large Language Models (LLMs) to build intelligent, production-ready AI systems.

Technical Skills

Programming Languages:

Python, SQL, JavaScript, XML

Machine Learning & AI:

Supervised Learning, Unsupervised Learning, Data Wrangling, Exploratory Data Analysis (EDA), Model Evaluation (Precision, Recall, F1-Score), Cross Validation, Hyperparameter Tuning, Feature Engineering

Natural Language Processing (NLP):

Tokenization, POS Tagging, Named Entity Recognition (NER), Dependency Parsing, Text Preprocessing, Regex-based NLP, Linguistic Rule Systems

Libraries & Frameworks:

Pandas, NumPy, Scikit-learn, SciPy, Matplotlib, Seaborn, Plotly, NLTK, SpaCy, FastAPI

Generative AI (Learning):

Large Language Models (LLMs), Prompt Engineering, RAG

Tools & Platforms:

Jupyter Notebook, VS Code, PyCharm, Git, Bitbucket, JIRA, Postman

Databases & Cloud:

Microsoft SQL Server, Azure (Basics)

Certifications

IBM Data Science Professional Certificate (Coursera – 12 course series)

Course Completion Year – 2026

EDUCATION

B.Tech – Computer Science & Engineering

Acharya College of Engineering Technology, Puducherry

CGPA: 8.36 | Year: 2018

LANGUAGES

Tamil (fluent), English (fluent), Malayalam (basic), French (basic)

PROFESSIONAL EXPERIENCE

Associate Programmer (NLP Engineer)

Integra Software Services Pvt. Ltd., Puducherry | Oct 2018 – Aug 2022

- Designed and implemented NLP pipelines for text processing, grammar analysis, and linguistic validation in a digital writing assistant.
- Developed and optimized 100+ NLP rules to improve text accuracy, grammar correction, and writing quality across multiple client applications.
- Improved system accuracy by 60% through iterative error analysis, rule optimization, and reduction of false positives/negatives.
- Automated unit testing processes using Python scripts, increasing test coverage and reducing manual QA efforts by 40%.
- Processed and analyzed large volumes of unstructured text data to enhance NLP model performance and efficiency.
- Collaborated in Agile development environments to deliver scalable, production-ready NLP solutions.
- Conducted onboarding and training sessions for new team members on NLP techniques, tools, and development workflows.

PROJECTS

1) AI Resume Analyser (using LLM):

- Developed a multi-agent resume screening system using LLM (GPT-4o-mini), prompt engineering, FastAPI, and Streamlit UI.
- Parsed and processed resume and job description documents using PyPDF2.
- Designed three agents for resume extraction, job description analysis, and candidate evaluation.
- Generated structured outputs including selection status, reasoning, and skills match score.

Project Github link: https://github.com/AnishaMohandass/AI_Resume_Analyser

2) SpaceX Falcon 9 Landing Prediction (Machine Learning Project)

- Built an end-to-end machine learning pipeline to predict Falcon 9 first-stage landing success using real-world data.
- Collected and processed data via SpaceX REST API and web scraping, followed by data cleaning and feature engineering.
- Performed exploratory data analysis using SQL and visualization tools to identify trends and patterns.
- Created interactive dashboards using Plotly Dash and Folium for to better understand launch patterns and outcomes.
- Developed multiple classification models including Logistic Regression, SVM, Decision Tree, and KNN and applied hyperparameter tuning using Grid Search to optimize model performance.
- Evaluated models using accuracy, precision, recall, and F1-score to identify the best-performing algorithm.

Project Github link: <https://github.com/AnishaMohandass/Applied-Data-Science-Capstone>

CAREER BREAK

Aug 2022 – Present

- Focused on upskilling in Data Science, Machine Learning, and AI.
- Completed industry-recognized certifications and hands-on projects.
- Strengthening knowledge in Generative AI concepts (RAG, LangGraph, Multi-Agent System).